

SOFTWARE CHANGE REQUEST

CR-009: Student Dashboard Course Display and Enrollment

1. Change Request Details

Field	Details
Change Request ID	CR-009
Title	Student Dashboard Course Display and Enrollment
Priority	High
Difficulty	Medium
Change Type	Frontend and Backend Enhancement
Affected Version	Student Management System v1.0 Baseline
Dependency	24ICT06

2. Business Requirement

The current Student Dashboard primarily displays the student's profile information, including registration number, full name, email, phone number, and member since date.

The Student Dashboard currently does not provide a direct workflow for displaying available courses and enrolling in a selected course.

The system shall therefore be enhanced to provide an Available Courses section on the Student Dashboard. The section shall display available course information and allow an authenticated student to enroll in a selected course.

The change shall integrate the course and enrollment functionality required for the student-facing workflow while maintaining the existing student authentication and profile functionality.

3. Current Behavior

Currently, the Student Dashboard displays the student's profile information, including:

- Registration Number
- Full Name
- Email
- Phone Number

- Member Since

The current Student Dashboard does not provide:

- An Available Courses section.
- Course details for students.
- An Enroll option.
- A direct student course enrollment workflow from the dashboard.

The existing student authentication and profile functionality shall remain unchanged.

4. Requested Change

The Student Dashboard shall be enhanced to include an Available Courses section.

The implementation shall:

- Continue displaying the existing student profile information.
- Provide available course information to authenticated students.
- Display relevant course details such as course title, category, level, duration, and price.
- Provide an Enroll button for available courses.
- Allow an authenticated student to enroll in a selected course.
- Process enrollment through the backend.
- Display an appropriate success message after successful enrollment.
- Display an appropriate error message when enrollment cannot be completed.
- Preserve the existing student authentication and authorization functionality.
- Avoid unnecessary modifications to unrelated student functionality.

5. Scope

5.1 In Scope

- Student Dashboard enhancement.
- Available Courses section.
- Course information display.
- Course selection.

- Student enrollment.
- Enrollment success and error messages.
- Required backend/API integration for course retrieval and enrollment.
- Responsive course display.
- Preservation of existing authentication and authorization.

5.2 Out of Scope

- Advanced course recommendation functionality.
- Payment gateway integration.
- Online course content.
- Certificate generation.
- Changes to unrelated student profile functionality.
- Changes to the authentication mechanism.
- Changes to the existing authorization mechanism.
- My Enrollments page modifications.

6. Dependency

The implementation depends on:

- Existing user authentication.
- Existing student information.
- Student authorization.
- A course data source.
- Course retrieval functionality.
- Enrollment processing functionality.
- Appropriate database storage for course and enrollment information.

Where required, the necessary course and enrollment backend components shall be introduced to support the requested student workflow.

7. Potentially Affected Files

The following files are expected to be affected or introduced depending on the final implementation structure.

Frontend

- frontend/src/pages/StudentDashboard.jsx
- frontend/src/services/api.js
- frontend/src/styles/studentDashboard.css

Backend

- backend/controllers/courseController.js
- backend/controllers/enrollmentController.js
- backend/routes/courseRoutes.js
- backend/routes/enrollmentRoutes.js

Database

- courses table
- enrollments table

8. Functional Requirements

FR-001: Available Courses

The Student Dashboard shall display available courses.

FR-002: Course Details

Students shall be able to view relevant details of available courses.

FR-003: Student Enrollment

Authenticated students shall be able to enroll in a selected course.

FR-004: Enrollment Confirmation

The system shall display a success message after successful enrollment.

FR-005: Enrollment Error Handling

The system shall display an appropriate error message when enrollment cannot be completed.

9. Security Requirements

SEC-001: Authentication

Only authenticated students shall be allowed to enroll in courses.

SEC-002: Authorization

Student authorization shall be enforced for enrollment operations.

SEC-003: Secure API Usage

Course and enrollment requests shall use the application's authenticated API mechanism.

SEC-004: Security Preservation

The implementation shall not bypass or weaken existing authentication and authorization controls.

10. Acceptance Criteria**AC-001**

Available courses are displayed on the Student Dashboard.

AC-002

Students can view the relevant details of available courses.

AC-003

An authenticated student can enroll in a selected course.

AC-004

Successful enrollment displays an appropriate confirmation message.

AC-005

Failed enrollment attempts display an appropriate error message.

AC-006

Existing student profile, authentication, and authorization functionality continue to work correctly after implementation.

11. Expected Outcome

After implementing CR-009, the Student Dashboard will continue to display the student's existing profile information and will additionally provide an Available Courses section.

Available courses will be displayed with relevant information such as course title, category, level, duration, and price.

Authenticated students will be able to select a course and enroll directly from the Student Dashboard. The system will display an appropriate success message when enrollment is completed successfully and an appropriate error message when enrollment cannot be completed.

The change will provide students with a direct course browsing and enrollment workflow while preserving the existing student profile, authentication, and authorization functionality.

12. Implementation Report Requirement

The Final Exam Software Change Implementation Report shall include:

- Implementation summary.
- Business requirement.
- Current behavior.
- Requested change.
- Scope and out-of-scope items.
- Dependencies.
- Affected files.
- Frontend implementation details.
- Backend implementation details.
- Database impact.
- Technical and security changes.
- Test cases and test results.
- Acceptance criteria verification.
- Screenshots and implementation evidence.
- Git branch and commit information.
- Final implementation status.
- Outstanding issues, if any.

